

SAGE

Scalable Agentic Grounded Evaluation
for Crop Disease Diagnosis

Muhammad Arbab Arshad, Tirtho Roy, Asheesh K. Singh,
Chinmay Hegde, Baskar Ganapathysubramanian, Arti Singh, Soumik Sarkar

Iowa State University New York University

The Problem

- Crop diseases reduce yields by **up to 40%**
- Existing datasets: limited crops, leaf-only, no symptom metadata
- Current classifiers: **black boxes** with no explanation
- No structured, verifiable symptom knowledge alongside images

What's missing?

- ✗ Broad crop coverage
- ✗ Multi-organ imagery
- ✗ Source-cited symptoms
- ✗ Transparent reasoning

What We Contribute



Large-Scale Dataset

~1.1M images across 53 crops and 259 disease classes.

Multi-organ coverage: leaf, stem, root, seed, ear, head.

Assembled from established benchmarks, expert-curated collections, and community sources.



Source-Cited Registry

Automated pipeline generates structured symptom knowledge for any crop.

Every field carries per-field provenance: value, source URL, and verbatim quote.

No manual curation required.



Agentic Diagnosis

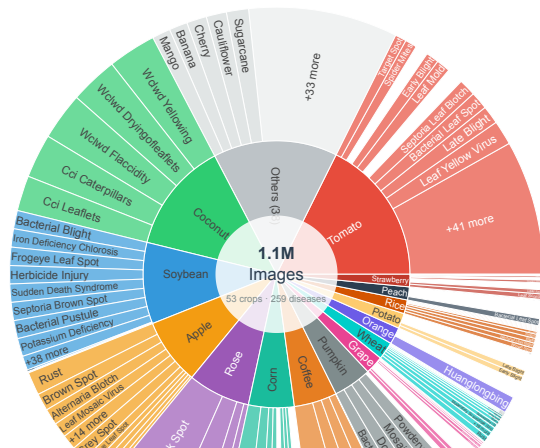
Each prediction made by an autonomous reasoning agent.

Guided chain of thought with anatomical narrowing and sequential reference comparison.

Full reasoning trace at every step.

No task-specific training. Works for any crop with reference images. Improves with better models.

Our Dataset



Standard dataset

Image + Disease label

That's it.

Standard dataset

Image + Disease label

That's it.

Ours

- Image + Disease label
- + **Organ tags** (leaf, stem, root, ...)
- + **Symptom descriptions**
- + **Source URL + Verbatim quote**

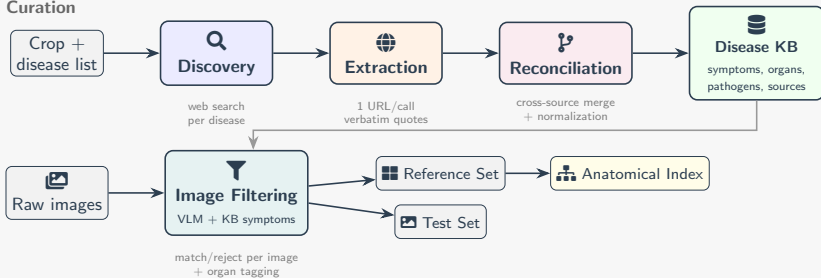
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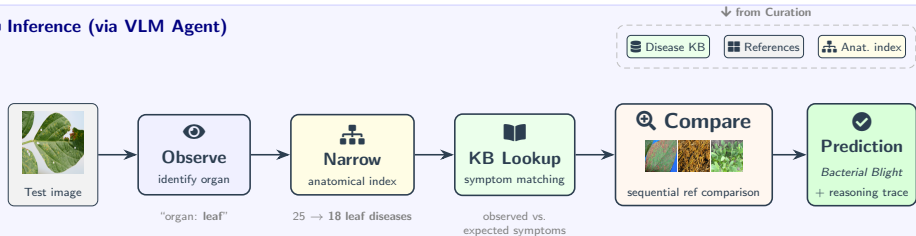
Example: Tomato Early Blight

System Overview

Curation



Inference (via VLM Agent)



Experimental Setup

- **Evaluation crops:** Soybean (25 classes), Corn (30 classes), Mango (4 classes)
- **Test images:** 3 per class (10 for Mango), fixed seed
- **Reference budget k :** 0, 1, 4, 8, 16 labeled images the agent may examine
- **KB source:** none (no symptoms) or internet (auto-generated registry)
- **Model tiers:** Haiku, Sonnet, Opus

Baseline

Agent at $k=0$, no KB.

The model receives only the test image and the list of candidate class names.

No reference images, no symptom descriptions.

Full pipeline

Agent + internet KB + k reference images.

Anatomical narrowing, symptom matching, sequential comparison.

Example: Agent Reasoning on Soybean



Test image

Prediction:
Bacterial Blight
Correct (0.82)

Sonnet, $k=4$, internet KB

Reasoning Trace

Step 1: Observe test image

Step 2: Dark brown/black irregular lesions with yellow halos, angular, concentrated along mid-vein. Green tissue between lesions intact.

Top candidates: Bacterial Blight, Septoria brown spot, Vein necrosis virus.

Step 3: View ref: Bacterial Blight → matches well, dark lesions + yellow-green halos.

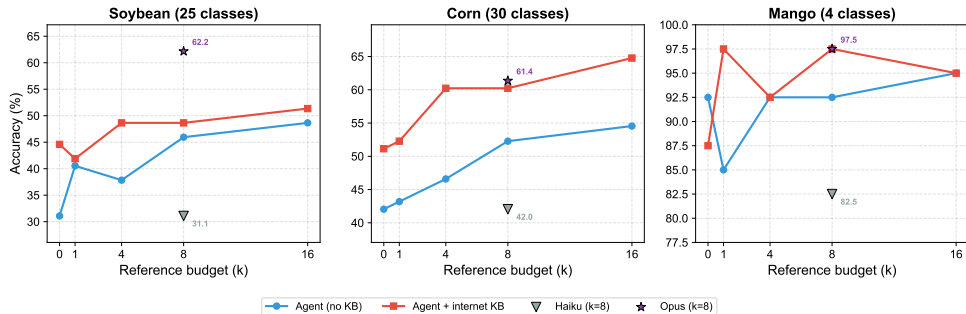
Step 4: View ref: Septoria brown spot → extensive yellowing, small dark spots. **Does not match.**

Step 5: View ref: Vein necrosis virus → diffuse yellow spots, no dark lesion centers. **Does not match.**

Step 6: View ref: Frogeye leaf spot → small circular lesions with holes. **Does not match.**

Conclusion: Yellow-green halos + dark angular lesions on otherwise green leaf = hallmark of **Bacterial Blight**.

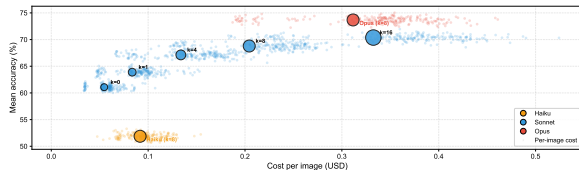
Results: Accuracy vs. Reference Budget



The knowledge base provides the largest gains at low reference budgets, where symptom descriptions and anatomical context guide the agent to examine the most relevant references first.

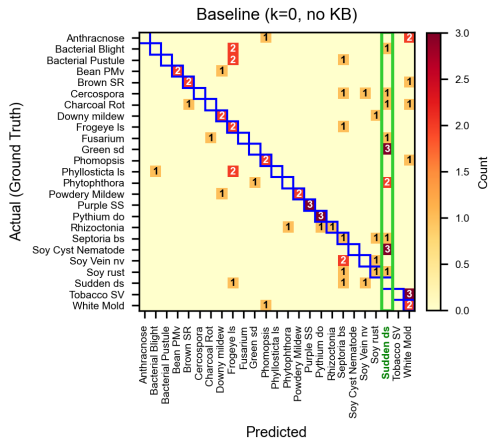
Mean improvement with KB: **+15.2 percentage points** at $k=16$.

Cost-Accuracy Tradeoff

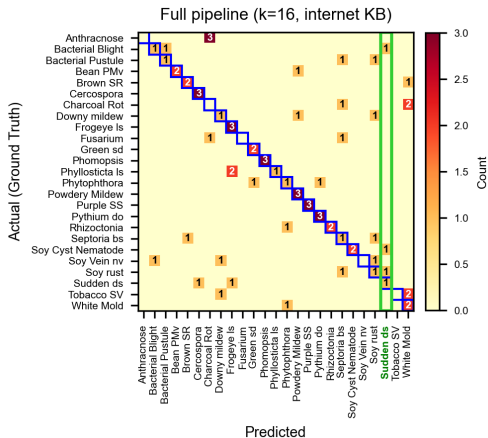


- Increasing k improves accuracy but at growing cost
- Model quality provides a complementary lever: at $k=8$, **Opus achieves higher accuracy than Sonnet at $k=16$ for comparable cost**
- The system improves directly as foundation models advance, **without retraining or architectural changes**

Impact: Reducing Over-Prediction (Soybean)



Baseline ($k=0$, no KB): 31.1%



Full pipeline ($k=16$, KB): 51.4%

Sudden death syndrome over-prediction: **14 false positives** → **3**.

The agent uses KB symptoms and reference comparisons to distinguish visually similar diseases.

Key Takeaways

What we built

- One of the largest plant disease image datasets (~1.1M images, 53 crops, 259 diseases)
- Automated pipeline that generates source-cited symptom knowledge for any crop
- Agentic diagnostic system where every prediction comes with a full, verifiable reasoning trace

What we found

- Symptom knowledge consistently improves accuracy across all crops
- The system improves with better foundation models, without retraining or architectural changes
- Reasoning traces allow users to verify diagnoses and identify failure points

Thank You

Questions?

Code and data will be made publicly available.